# Learning Day 11
# Hyperparameter Tuning

BI presentation from the main MVD notebook and extended source-coverage notebook

**Core decision**

Tuning improves model quality, but complexity must earn its place. For Santander, tuned XGBoost beat neural networks on the most decision-relevant metrics.

**Prepared for university submission**
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# Executive Summary

The project tested classical tuning, Bayesian tuning, neural architecture search, and extended-source methods.

| 0.9316 | 0.8733 | 0.9971 | 0 errors |
|---|---|---|---|
| **Bank best ROC-AUC** | **Santander best ROC-AUC** | **Extended best ROC-AUC** | **Execution QA** |
| RandomizedSearch RF | Bayesian XGBoost | Logistic/CASH | Refined notebook |

### What changed after tuning?

Bank Marketing: RandomizedSearch RF improved F1-score to 0.602 and ROC-AUC to 0.932. Santander: Bayesian Search delivered best ROC-AUC of 0.873. Neural networks improved learning coverage, but not the final business decision versus tuned XGBoost.

### Executive conclusion

For tabular business data, hyperparameter tuning added measurable value. However, model complexity must justify itself. The final recommendation is XGBoost for Santander and RandomizedSearch Random Forest for Bank Marketing.

### Submission story

The main notebook covers the university assignment workflow. The extended notebook closes the uploaded-source coverage gaps with KNN, Logistic Regression, Ridge, CASH, Hyperopt, Optuna, Successive Halving, and NAS theory.

# Two-Notebook Submission Architecture

The final submission separates the MVD project flow from the extended source-coverage appendix.

### Main notebook: hypertuning.ipynb

Purpose: Complete Day 11 MVD workflow. Datasets: Bank Marketing and Santander. Methods: Random Forest, XGBoost, KerasTuner neural networks. Outputs: Separate dataset charts, convergence chart, final recommendation.

### Extended notebook: source coverage

Purpose: Close uploaded-source gaps missed by MVD. Datasets: Stable local scikit-learn datasets. Methods: KNN, Logistic Regression, Ridge, CASH, Hyperopt, Optuna, Successive Halving, NAS theory. Outputs: Executed local-safe workflow with no error outputs.

### Data storytelling note

The main notebook answers the university assignment directly. The extended notebook proves broader source coverage and adds complete practical workflows for additional tuning methods. Together, they form the full Day 11 submission.

# Dataset Strategy

Different datasets were used because each tuning section needed a different modelling problem.

| Dataset | Task | Used for | Business learning value |
| --- | --- | --- | --- |
| Bank Marketing | Classification | Random Forest tuning | Marketing campaign response prediction |
| Santander | Classification | XGBoost, Bayesian, NN | High-dimensional customer transaction prediction |
| Breast Cancer | Classification | Extended methods | Safe local tuning comparison |
| Diabetes | Regression | Ridge regression | Regression HPO coverage |

**Important**

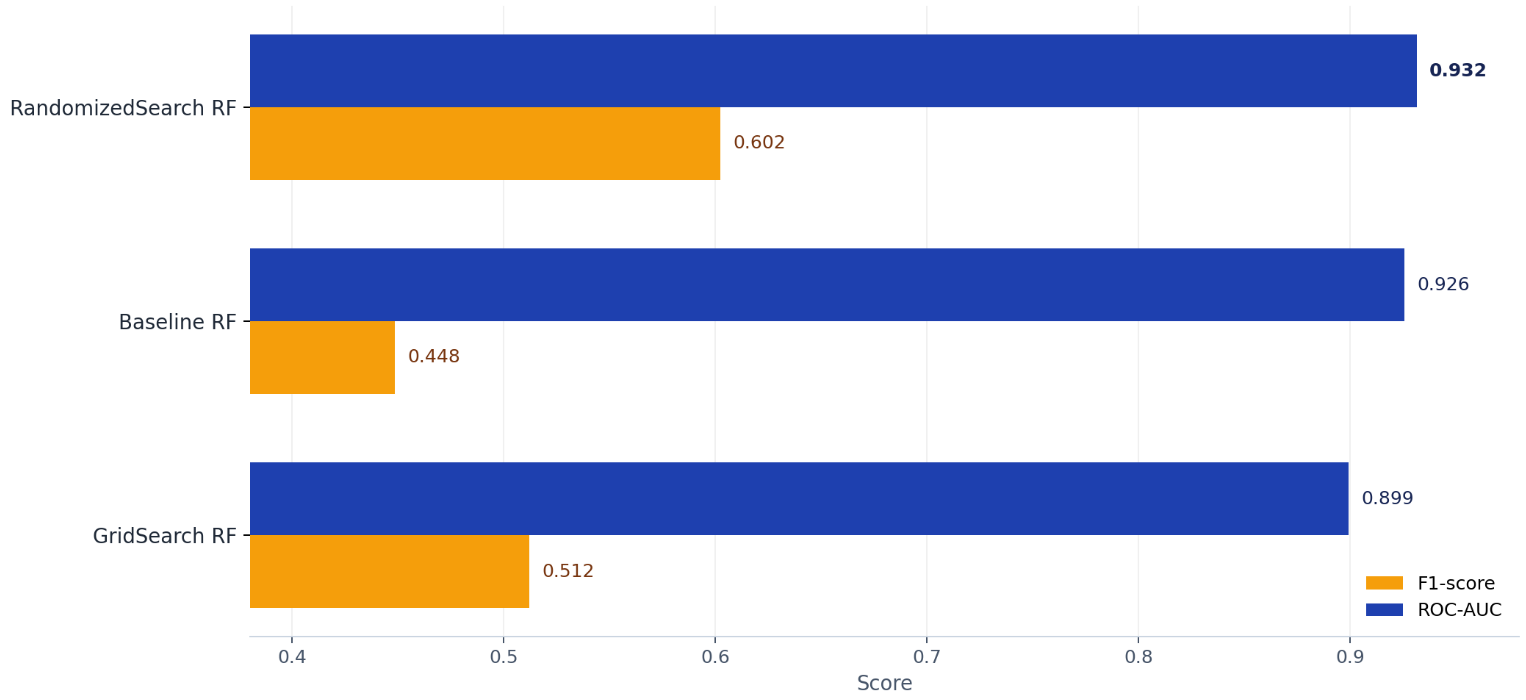
Results are compared within each dataset, not across different datasets. This avoids false conclusions between different business problems.

# Bank Marketing: Tuning Improved Positive-Class Detection

RandomizedSearch RF became the strongest Bank Marketing option.

## Bank Marketing - Tuning Changed the Detection Story

Test-set F1-score and ROC-AUC for Random Forest tuning strategies




RandomizedSearch RF delivered the best balance: highest F1-score (0.602) and highest ROC-AUC (0.932).

### Data story

The baseline already ranked customers well, but RandomizedSearch improved F1-score strongly. This matters because the positive response class is harder to detect.

**0.602**

**Best F1-score**

RandomizedSearch RF

**0.932**

**Best ROC-AUC**

RandomizedSearch RF

### Recommendation

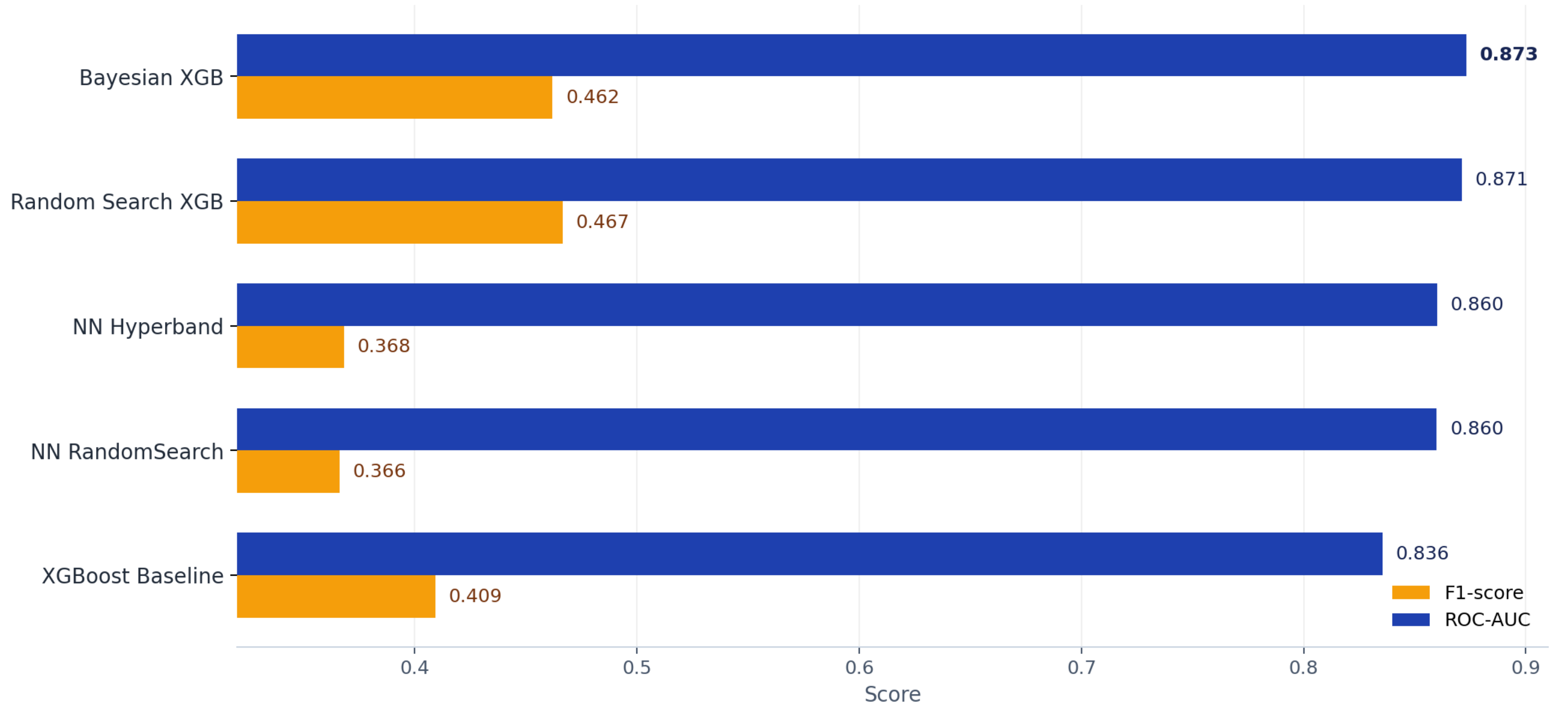
Use RandomizedSearch RF for the Bank Marketing model because it has the strongest F1-score and strongest ROC-AUC in this dataset.

# Santander: Complexity Did Not Beat Tuned XGBoost

High accuracy from neural networks was misleading; XGBoost won on business metrics.

## Santander - XGBoost Beat Neural Network Complexity

Final test-set comparison after XGBoost tuning, KerasTuner RandomSearch, and Hyperband




Bayesian XGBoost had the best ROC-AUC (0.873), while Random Search XGBoost had the best F1-score (0.467).

### Insight

Neural networks had higher accuracy, but lower F1-score and ROC-AUC. On an imbalanced dataset, accuracy alone is not enough.

**0.873**
**Best AUC**
Bayesian XGB

**0.467**
**Best F1**
Random XGB

### Business decision

Choose XGBoost over neural networks for Santander. It provides stronger ranking and detection quality with less operational complexity.

# Complexity Test: Does Deep Learning Earn Its Place?

The answer for this Santander dataset is no.

**0.8733**
**Best XGBoost ROC-AUC**
Bayesian tuned XGB

**0.8601**
**Best NN ROC-AUC**
Hyperband NN

**0.4666**
**Best XGBoost F1**
Random Search XGB

**0.3682**
**Best NN F1**
Hyperband NN

### What happened?

KerasTuner successfully tuned neural-network architectures, and Hyperband improved efficiency. But the final neural-network metrics did not beat tuned XGBoost, especially on F1-score and ROC-AUC.

### Decision rule

A more complex model is justified only when it produces better business metrics or materially improves stability, bias, runtime, cost savings, or explainability. Here, XGBoost remained stronger.

### Final complexity decision

Deep learning belongs in the notebook because it was required and valuable for learning. It does not belong as the recommended production model for this Santander dataset.

# Search Strategy: Guided Search vs Blind Search

Better validation score is useful, but final model choice depends on test metrics.

## Search Strategy - Better Search Is Not Always Faster

Tuning quality must be judged together with runtime and final test-set behavior


Bayesian XGBoost had the strongest CV AUC, but Random Search XGBoost delivered the strongest Santander F1-score on test data.

### Key message

Bayesian Search used previous trials to guide future trials and achieved the best CV AUC for XGBoost.

### Practical lesson

Search quality must be judged by final test performance, not only validation or CV score.

### Runtime tradeoff

More advanced search is not automatically faster. It must be justified by better performance or better business value.

# Why the Extended Notebook Was Needed

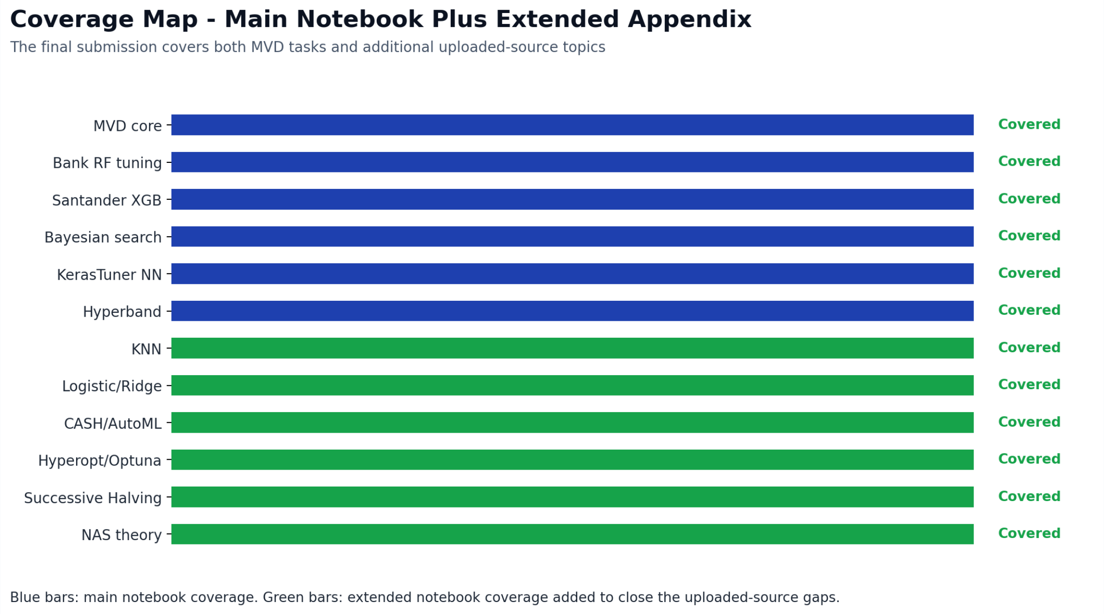
The main notebook completed the assignment, but not the full uploaded-source scope.

## Gap after the main notebook

Missing practical topics: KNN, Logistic Regression, Ridge, Hyperopt, Optuna, CASH, Successive Halving, and deeper NAS/Bayesian theory.

## Appendix solution

Stable local datasets were used to cover all missed methods without internet downloads, long code cells, or unstable deep-learning installs.

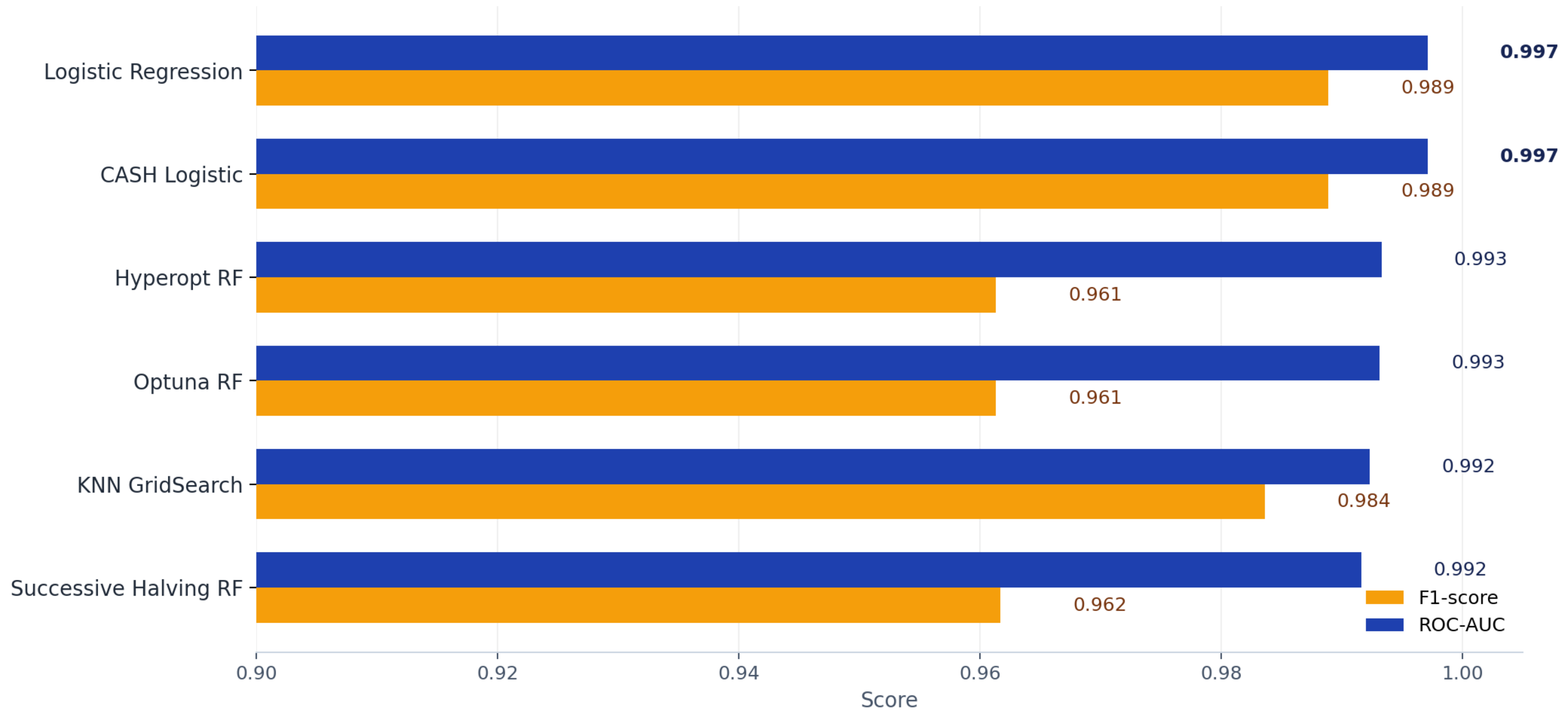


# Extended Notebook Results: Interpretable Models Dominated

The appendix demonstrated full practical coverage using stable local data.

## Extended Notebook - Source Coverage Models Worked End-to-End

Classification appendix results using local-safe datasets and stable workflows




Logistic Regression and CASH selected the same strong, interpretable model: ROC-AUC 0.997 and F1-score 0.989.

**0.9971**
**Top ROC-AUC**
Logistic/CASH

**0.9889**
**Top F1**
Logistic/CASH

### Interpretation

The appendix proves that simple, interpretable models can perform extremely well when the dataset is clean and well prepared.

### Submission value

This section closes uploaded-source coverage for KNN, Logistic Regression, CASH, Hyperopt, Optuna, and Successive Halving.

# Regression, CASH, Hyperopt, Optuna, and Multi-Fidelity Coverage

The appendix expands Day 11 beyond classification-only tuning.

| 53.20 | 0.488 | 0.9933 | 0.9931 |
|---|---|---|---|
| **Ridge RMSE** | **Ridge R-squared** | **Hyperopt RF AUC** | **Optuna RF AUC** |
| Regression tuning | Explained variance | Search coverage | Modern HPO |

## Regression tuning

Ridge Regression closed the regression-side tuning workflow using alpha selection by GridSearchCV, followed by test RMSE, MAE, and R-squared interpretation.

## AutoML-style tuning

CASH compares algorithm families, Hyperopt and Optuna show modern automated search workflows, and Successive Halving adds multi-fidelity resource control.

## Why this matters

The final submission now proves that hyperparameter tuning was understood as a broad model-selection discipline, not just GridSearchCV and RandomizedSearchCV.

# Final Model Decisions

The final model choice depends on business objective, not only maximum accuracy.

### Bank Marketing

Use RandomizedSearch Random Forest. It delivered the best Bank F1-score and strongest ROC-AUC improvement.

### Santander

Use Bayesian Tuned XGBoost for ranking quality. Use Random Search XGBoost when F1-score is the priority.

### Extended coverage

Use the appendix as source-proof. It covers KNN, Logistic Regression, Ridge, CASH, Hyperopt, Optuna, Successive Halving, and NAS.

### Core business message

Hyperparameter tuning is valuable, but it should be governed by metric relevance, model complexity, runtime, and explainability. The best-performing approach is not always the newest or most complex one.

### Recommended final story for submission

Start with main notebook results, then use the extended notebook to show broader source mastery and robust local execution.

# Submission QA and Risk Control

The final artefacts were designed to avoid local execution and submission risks.

**91**
**Refined notebook cells**
Extended notebook

**24**
**Code cells**
Short and commented

**0**
**Error outputs**
Execution-safe

**0**
**Inline installs**
Kernel-safe

## Risk controls

No internet dependency in the extended notebook. No TensorFlow or KerasTuner in the local-safe appendix. Small code cells with comments. Stable built-in datasets and output-based conclusions.

## Presentation QA

Wide slide layout, dark-style headers, readable charts, separate dataset views, no mixed-dataset comparison chart, and clear business storytelling.

## Submission control

The final PDF presentation should be submitted together with the notebooks and the individual documentation PDFs.

# Final Takeaway

The submission proves the Day 11 assignment and the broader uploaded-source topics. The strongest data story is simple: tuning helps, but the best model is the one that improves the right metric without unnecessary complexity.

**Recommended final answer to university**

For the Santander business case, Bayesian/Randomized XGBoost is stronger than the tuned neural-network options. For source coverage and theory, the extended notebook completes the wider HPO and AutoML topics.